

Package ‘fitVARMxID’

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Title Fit the Vector Autoregressive Model for Multiple Individuals

Version 1.0.2

Description Fit the vector autoregressive model for multiple individuals using the 'OpenMx' package (Hunter, 2017 <[doi:10.1080/10705511.2017.1369354](https://doi.org/10.1080/10705511.2017.1369354)>).

URL <https://github.com/jeksterslab/fitVARMxID>,
<https://jeksterslab.github.io/fitVARMxID/>

BugReports <https://github.com/jeksterslab/fitVARMxID/issues>

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Contents

coef.varmxid	2
converged	4
FitVARMxID	5
LDL	14
print.varmxid	15
Softplus	17
summary.varmxid	18
vcov.varmxid	20

Index	23
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coef.varmxid	<i>Parameter Estimates</i>
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Description

Parameter Estimates

Usage

```
## S3 method for class 'varmxid'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  var_metric = "var",
  converged = TRUE,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  prop = FALSE,
  grad_tol = 0.01,
  hess_tol_abs = 1e-08,
  hess_tol_rel = 1e-10,
  check_condition = FALSE,
  cond_max = 1e+12,
  abs_bnd_tol = 1e-06,
  rel_bnd_tol = 1e-04,
  ok_codes = 0L,
  require_finite_fit = TRUE,
  ...
)
```

Arguments

object	Object of class varmxid.
mu	Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.

psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
var_metric	Character string. If var_metric = "var", psi and theta are in the original variance/covariance metric. If var_metric = "logvar", the diagonal elements of psi and theta are the log of the variances and the off-diagonal elements correspond to strict L in the LDL ' decomposition. If var_metric = "softplusvar", the diagonal elements of psi and theta are the softplus of the variances and the off-diagonal elements correspond to strict L in the LDL ' decomposition.
converged	Logical. Only include converged cases.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
prop	Logical. If prop = FALSE, a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If prop = TRUE, the proportion of cases that converged.
grad_tol	Numeric. Tolerance for the gradient-based convergence check.
hess_tol_abs	Numeric. Absolute tolerance used when checking whether the Hessian is positive-definite.
hess_tol_rel	Numeric. Relative tolerance used when checking whether the Hessian is positive-definite.
check_condition	Logical. If TRUE, also check the Hessian condition number as part of the Hessian diagnostics.
cond_max	Numeric. Maximum allowed condition number for the Hessian when check_condition = TRUE.
abs_bnd_tol	Numeric. Absolute tolerance for detecting whether a parameter is at its lower or upper bound.
rel_bnd_tol	Numeric. Relative tolerance for detecting whether a parameter is at its lower or upper bound.
ok_codes	Integer vector. Optimizer status codes that are considered acceptable.
require_finite_fit	Logical. If TRUE, require a finite objective value (fit) as part of the convergence check.
...	additional arguments.

Value

Returns a list of vectors of parameter estimates.

Author(s)

Ivan Jacob Agaloos Pesigan

converged

*Check Model Convergence***Description**

Determines whether each fitted **OpenMx** model in a `varmxid` object meets convergence criteria based on (a) acceptable optimizer status and gradient size, (b) a positive-definite Hessian, (c) parameters not being at their bounds, and (optionally) non-vanishing measurement error variances (theta).

Usage

```
converged(object, ...)

## S3 method for class 'varmxid'
converged(
  object,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  prop = FALSE,
  grad_tol = 0.01,
  hess_tol_abs = 1e-08,
  hess_tol_rel = 1e-10,
  check_condition = FALSE,
  cond_max = 1e+12,
  abs_bnd_tol = 1e-06,
  rel_bnd_tol = 1e-04,
  ok_codes = 0L,
  require_finite_fit = TRUE,
  ...
)
```

Arguments

<code>object</code>	A fit object.
<code>...</code>	Passed to and/or used by methods.
<code>vanishing_theta</code>	Logical. Test for measurement error variance going to zero if converged = TRUE.
<code>theta_tol</code>	Numeric. Tolerance for vanishing theta test if converged and <code>theta_tol</code> are TRUE.
<code>prop</code>	Logical. If <code>prop = FALSE</code> , a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If <code>prop = TRUE</code> , the proportion of cases that converged.
<code>grad_tol</code>	Numeric. Tolerance for the gradient-based convergence check.

<code>hess_tol_abs</code>	Numeric. Absolute tolerance used when checking whether the Hessian is positive-definite.
<code>hess_tol_rel</code>	Numeric. Relative tolerance used when checking whether the Hessian is positive-definite.
<code>check_condition</code>	Logical. If TRUE, also check the Hessian condition number as part of the Hessian diagnostics.
<code>cond_max</code>	Numeric. Maximum allowed condition number for the Hessian when <code>check_condition</code> = TRUE.
<code>abs_bnd_tol</code>	Numeric. Absolute tolerance for detecting whether a parameter is at its lower or upper bound.
<code>rel_bnd_tol</code>	Numeric. Relative tolerance for detecting whether a parameter is at its lower or upper bound.
<code>ok_codes</code>	Integer vector. Optimizer status codes that are considered acceptable.
<code>require_finite_fit</code>	Logical. If TRUE, require a finite objective value (fit) as part of the convergence check.

Value

For the `varmxid` method: If `prop` = FALSE, a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If `prop` = TRUE, the proportion of cases that converged.

Author(s)

Ivan Jacob Agaloos Pesigan

FitVARMxID

Fit the First-Order Vector Autoregressive Model by ID

Description

The function fits the first-order vector autoregressive model for each unit ID.

Usage

```
FitVARMxID(
  data,
  observed,
  id,
  time = NULL,
  ct = FALSE,
  center = TRUE,
  mu_fixed = FALSE,
  mu_free = NULL,
  mu_values = NULL,
```

```
mu_lbound = NULL,  
mu_ubound = NULL,  
alpha_fixed = FALSE,  
alpha_free = NULL,  
alpha_values = NULL,  
alpha_lbound = NULL,  
alpha_ubound = NULL,  
beta_fixed = FALSE,  
beta_free = NULL,  
beta_values = NULL,  
beta_lbound = NULL,  
beta_ubound = NULL,  
psi_diag = FALSE,  
psi_fixed = FALSE,  
psi_d_free = NULL,  
psi_d_values = NULL,  
psi_d_lbound = NULL,  
psi_d_ubound = NULL,  
psi_d_equal = FALSE,  
psi_l_free = NULL,  
psi_l_values = NULL,  
psi_l_lbound = NULL,  
psi_l_ubound = NULL,  
nu_fixed = TRUE,  
nu_free = NULL,  
nu_values = NULL,  
nu_lbound = NULL,  
nu_ubound = NULL,  
theta_diag = TRUE,  
theta_fixed = TRUE,  
theta_d_free = NULL,  
theta_d_values = NULL,  
theta_d_lbound = NULL,  
theta_d_ubound = NULL,  
theta_d_equal = FALSE,  
theta_l_free = NULL,  
theta_l_values = NULL,  
theta_l_lbound = NULL,  
theta_l_ubound = NULL,  
mu0_fixed = TRUE,  
mu0_func = TRUE,  
mu0_free = NULL,  
mu0_values = NULL,  
mu0_lbound = NULL,  
mu0_ubound = NULL,  
sigma0_fixed = TRUE,  
sigma0_func = TRUE,  
sigma0_diag = FALSE,
```

```

sigma0_d_free = NULL,
sigma0_d_values = NULL,
sigma0_d_lbound = NULL,
sigma0_d_ubound = NULL,
sigma0_d_equal = FALSE,
sigma0_l_free = NULL,
sigma0_l_values = NULL,
sigma0_l_lbound = NULL,
sigma0_l_ubound = NULL,
robust = FALSE,
seed = NULL,
tries_explore = 100,
tries_local = 100,
max_attempts = 10,
grad_tol = 0.01,
hess_tol_abs = 1e-08,
hess_tol_rel = 1e-10,
check_condition = FALSE,
cond_max = 1e+12,
abs_bnd_tol = 1e-06,
rel_bnd_tol = 1e-04,
ok_codes = 0L,
require_finite_fit = TRUE,
silent = FALSE,
ncores = NULL
)

```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), and at least one column of observed values.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data. Used when ct = TRUE.
ct	Logical. If TRUE, fit a continuous-time vector autoregressive model. If FALSE, fit a discrete-time vector autoregressive model.
center	Logical. If TRUE, use the mean-centered (mean-reverting) state equation. When center = TRUE, alpha is implied and the set-point mu is estimated. When center = FALSE, alpha is estimated and mu is implied.
mu_fixed	Logical. If TRUE, the set-point mean vector mu is fixed to mu_values. If mu_fixed = TRUE and mu_values = NULL, mu is fixed to a zero vector. If FALSE, mu is estimated.
mu_free	Logical vector indicating which elements of mu are freely estimated. If NULL, all elements are free. Ignored if mu_fixed = TRUE.

mu_values	Numeric vector of values for mu. If mu_fixed = TRUE, these are fixed values. If mu_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.
mu_lbound	Numeric vector of lower bounds for mu. Ignored if mu_fixed = TRUE.
mu_ubound	Numeric vector of upper bounds for mu. Ignored if mu_fixed = TRUE.
alpha_fixed	Logical. If TRUE, the dynamic model intercept vector alpha is fixed to alpha_values. If FALSE, alpha is estimated.
alpha_free	Logical vector indicating which elements of alpha are freely estimated. If NULL, all elements are free. Ignored if alpha_fixed = TRUE.
alpha_values	Numeric vector of values for alpha. If alpha_fixed = TRUE, these are fixed values. If alpha_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.
alpha_lbound	Numeric vector of lower bounds for alpha. Ignored if alpha_fixed = TRUE.
alpha_ubound	Numeric vector of upper bounds for alpha. Ignored if alpha_fixed = TRUE.
beta_fixed	Logical. If TRUE, the dynamic model coefficient matrix beta is fixed. If FALSE, beta is estimated.
beta_free	Logical matrix indicating which elements of beta are freely estimated. If NULL, all elements are free. Ignored if beta_fixed = TRUE.
beta_values	Numeric matrix. Values for beta. If beta_fixed = TRUE, these are fixed values; if beta_fixed = FALSE, these are starting values. If NULL, defaults to a diagonal matrix with -0.001 when ct = TRUE and 0.001 when ct = FALSE.
beta_lbound	Numeric matrix of lower bounds for beta. If NULL, defaults to -2.5. Ignored if beta_fixed = TRUE.
beta_ubound	Numeric matrix. Upper bounds for beta. Ignored if beta_fixed = TRUE. If NULL, defaults to +2.5. If NULL and ct = TRUE, diagonal upper bounds are set to -1e-05.
psi_diag	Logical. If TRUE, psi is diagonal. If FALSE, psi is symmetric.
psi_fixed	Logical. If TRUE, the process noise covariance matrix psi is fixed using psi_d_values and psi_l_values. If psi_d_values is NULL it is fixed to a zero matrix. If FALSE, psi is estimated.
psi_d_free	Logical vector indicating free/fixed status of the elements of psi_d. If NULL, all element of psi_d are free.
psi_d_values	Numeric vector with starting values for psi_d. If psi_fixed = TRUE, these are fixed values. If psi_fixed = FALSE, these are starting values.
psi_d_lbound	Numeric vector with lower bounds for psi_d.
psi_d_ubound	Numeric vector with upper bounds for psi_d.
psi_d_equal	Logical. When TRUE, all free diagonal elements of psi_d are constrained to be equal and estimated as a single shared parameter. Ignored if no diagonal elements are free.
psi_l_free	Logical matrix indicating which strictly-lower-triangular elements of psi_l are free. If NULL, all element of psi_l are free. Ignored if psi_diag = TRUE.
psi_l_values	Numeric matrix of starting values for the strictly-lower-triangular elements of psi_l.

psi_l_lbound	Numeric matrix with lower bounds for psi_l.
psi_l_ubound	Numeric matrix with upper bounds for psi_l.
nu_fixed	Logical. If TRUE, the measurement model intercept vector nu is fixed to nu_values. If FALSE, nu is estimated.
nu_free	Logical vector indicating which elements of nu are freely estimated. If NULL, all elements are free. Ignored if nu_fixed = TRUE.
nu_values	Numeric vector of values for nu. If nu_fixed = TRUE, these are fixed values. If nu_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros.
nu_lbound	Numeric vector of lower bounds for nu. Ignored if nu_fixed = TRUE.
nu_ubound	Numeric vector of upper bounds for nu. Ignored if nu_fixed = TRUE.
theta_diag	Logical. If TRUE, theta is diagonal. If FALSE, theta is symmetric.
theta_fixed	Logical. If TRUE, the measurement error covariance matrix theta is fixed using theta_d_values and theta_l_values. If theta_d_values is NULL it is fixed to a zero matrix. If FALSE, theta is estimated.
theta_d_free	Logical vector indicating free/fixed status of the diagonal parameters theta_d. If NULL, all element of theta_d are free.
theta_d_values	Numeric vector with starting values for theta_d. If theta_fixed = TRUE, these are fixed values. If theta_fixed = FALSE, these are starting values.
theta_d_lbound	Numeric vector with lower bounds for theta_d.
theta_d_ubound	Numeric vector with upper bounds for theta_d.
theta_d_equal	Logical. When TRUE, all free diagonal elements of theta_d are constrained to be equal and estimated as a single shared parameter. Ignored if no diagonal elements are free.
theta_l_free	Logical matrix indicating which strictly-lower-triangular elements of theta_l are free. If NULL, all element of theta_l are free. Ignored if theta_diag = TRUE.
theta_l_values	Numeric matrix of starting values for the strictly-lower-triangular elements of theta_l.
theta_l_lbound	Numeric matrix with lower bounds for theta_l.
theta_l_ubound	Numeric matrix with upper bounds for theta_l.
mu0_fixed	Logical. If TRUE, the initial mean vector mu0 is fixed. If mu0_fixed = TRUE and mu0_func = TRUE, mu0 is fixed to the implied stable mean vector. If mu0_fixed = TRUE and mu0_values = NULL, mu0 is fixed to a zero vector. If FALSE, mu0 is estimated.
mu0_func	Logical. If TRUE and mu0_fixed = TRUE, mu0 is fixed to the implied stable mean vector.
mu0_free	Logical vector indicating which elements of mu0 are free. Ignored if mu0_fixed = TRUE.
mu0_values	Numeric vector of values for mu0. If mu0_fixed = TRUE, these are fixed values. If mu0_fixed = FALSE, these are starting values. If NULL, defaults to a vector of zeros. Ignored if mu0_fixed = TRUE and mu0_func = TRUE.

<code>mu0_lbound</code>	Numeric vector of lower bounds for <code>mu0</code> . Ignored if <code>mu0_fixed = TRUE</code> .
<code>mu0_ubound</code>	Numeric vector of upper bounds for <code>mu0</code> . Ignored if <code>mu0_fixed = TRUE</code> .
<code>sigma0_fixed</code>	Logical. If <code>TRUE</code> , the initial condition covariance matrix <code>sigma0</code> is fixed using <code>sigma0_d_values</code> and <code>sigma0_l_values</code> . If <code>sigma0_fixed = TRUE</code> and <code>sigma0_func = TRUE</code> , <code>sigma0</code> is fixed to the implied stable covariance matrix. If <code>sigma0_fixed = TRUE</code> and <code>sigma0_d_values = NULL</code> , <code>sigma0</code> is fixed to a diffused matrix.
<code>sigma0_func</code>	Logical. If <code>TRUE</code> and <code>sigma0_fixed = TRUE</code> , <code>sigma0</code> is fixed to the implied stable covariance matrix.
<code>sigma0_diag</code>	Logical. If <code>TRUE</code> , <code>sigma0</code> is diagonal. If <code>FALSE</code> , <code>sigma0</code> is symmetric.
<code>sigma0_d_free</code>	Logical vector indicating free/fixed status of the elements of <code>sigma0_d</code> . If <code>NULL</code> , all element of <code>sigma0_d</code> are free.
<code>sigma0_d_values</code>	Numeric vector with starting values for <code>sigma0_d</code> . If <code>sigma0_fixed = TRUE</code> , these are fixed values. If <code>sigma0_fixed = FALSE</code> , these are starting values.
<code>sigma0_d_lbound</code>	Numeric vector with lower bounds for <code>sigma0_d</code> .
<code>sigma0_d_ubound</code>	Numeric vector with upper bounds for <code>sigma0_d</code> .
<code>sigma0_d_equal</code>	Logical. When <code>TRUE</code> , all free diagonal elements of <code>sigma0_d</code> are constrained to be equal and estimated as a single shared parameter. Ignored if no diagonal elements are free.
<code>sigma0_l_free</code>	Logical matrix indicating which strictly-lower-triangular elements of <code>sigma0_l</code> are free. If <code>NULL</code> , all element of <code>sigma0_l</code> are free. Ignored if <code>sigma0_diag = TRUE</code> .
<code>sigma0_l_values</code>	Numeric matrix of starting values for the strictly-lower-triangular elements of <code>sigma0_l</code> .
<code>sigma0_l_lbound</code>	Numeric matrix with lower bounds for <code>sigma0_l</code> .
<code>sigma0_l_ubound</code>	Numeric matrix with upper bounds for <code>sigma0_l</code> .
<code>robust</code>	Logical. If <code>TRUE</code> , calculate robust (sandwich) sampling variance-covariance matrix.
<code>seed</code>	Random seed for reproducibility.
<code>tries_explore</code>	Integer. Number of extra tries for the wide exploration phase.
<code>tries_local</code>	Integer. Number of extra tries for local polishing.
<code>max_attempts</code>	Integer. Maximum number of remediation attempts after the first Hessian computation fails the criteria.
<code>grad_tol</code>	Numeric. Tolerance for the gradient-based convergence check.
<code>hess_tol_abs</code>	Numeric. Absolute tolerance used when checking whether the Hessian is positive-definite.
<code>hess_tol_rel</code>	Numeric. Relative tolerance used when checking whether the Hessian is positive-definite.

check_condition	Logical. If TRUE, also check the Hessian condition number as part of the Hessian diagnostics.
cond_max	Numeric. Maximum allowed condition number for the Hessian when check_condition = TRUE.
abs_bnd_tol	Numeric. Absolute tolerance for detecting whether a parameter is at its lower or upper bound.
rel_bnd_tol	Numeric. Relative tolerance for detecting whether a parameter is at its lower or upper bound.
ok_codes	Integer vector. Optimizer status codes that are considered acceptable.
require_finite_fit	Logical. If TRUE, require a finite objective value (fit) as part of the convergence check.
silent	Logical. If TRUE, suppresses messages during the model fitting stage.
ncores	Positive integer. Number of cores to use.

Details

Measurement Model:

By default, the measurement model is given by

$$\mathbf{y}_{i,t} = \boldsymbol{\eta}_{i,t}.$$

However, the full measurement model can be parameterized as follows

$$\mathbf{y}_{i,t} = \boldsymbol{\nu}_i + \boldsymbol{\Lambda} \boldsymbol{\eta}_{i,t} + \boldsymbol{\varepsilon}_{i,t}, \quad \text{with} \quad \boldsymbol{\varepsilon}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}_i)$$

where $\mathbf{y}_{i,t}$, $\boldsymbol{\eta}_{i,t}$, and $\boldsymbol{\varepsilon}_{i,t}$ are random variables and $\boldsymbol{\nu}_i$, $\boldsymbol{\Lambda}$, and $\boldsymbol{\Theta}_i$ are model parameters. $\mathbf{y}_{i,t}$ represents a vector of observed random variables, $\boldsymbol{\eta}_{i,t}$ a vector of latent random variables, and $\boldsymbol{\varepsilon}_{i,t}$ a vector of random measurement errors, at time t and individual i . $\boldsymbol{\nu}_i$ denotes a vector of intercepts (fixed to a null vector by default), $\boldsymbol{\Lambda}$ a matrix of factor loadings, and $\boldsymbol{\Theta}_i$ the covariance matrix of $\boldsymbol{\varepsilon}$. In this model, $\boldsymbol{\Lambda}$ is an identity matrix and $\boldsymbol{\Theta}_i$ is a diagonal matrix.

Discrete-Time Dynamic Structure:

The dynamic structure is given by

$$\boldsymbol{\eta}_{i,t} = \boldsymbol{\alpha}_i + \boldsymbol{\beta}_i \boldsymbol{\eta}_{i,t-1} + \boldsymbol{\zeta}_{i,t}, \quad \text{with} \quad \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}_i)$$

where $\boldsymbol{\eta}_{i,t}$, $\boldsymbol{\eta}_{i,t-1}$, and $\boldsymbol{\zeta}_{i,t}$ are random variables, and $\boldsymbol{\alpha}_i$, $\boldsymbol{\beta}_i$, and $\boldsymbol{\Psi}_i$ are model parameters. Here, $\boldsymbol{\eta}_{i,t}$ is a vector of latent variables at time t and individual i , $\boldsymbol{\eta}_{i,t-1}$ represents a vector of latent variables at time $t-1$ and individual i , and $\boldsymbol{\zeta}_{i,t}$ represents a vector of dynamic noise at time t and individual i . $\boldsymbol{\alpha}_i$ denotes a vector of intercepts, $\boldsymbol{\beta}_i$ a matrix of autoregression and cross regression coefficients, and $\boldsymbol{\Psi}_i$ the covariance matrix of $\boldsymbol{\zeta}_{i,t}$.

If center = TRUE, the dynamic structure is parameterized as follows

$$\boldsymbol{\eta}_{i,t} = \boldsymbol{\mu}_i + \boldsymbol{\beta}_i (\boldsymbol{\eta}_{i,t-1} - \boldsymbol{\mu}_i) + \boldsymbol{\zeta}_{i,t}$$

where $\boldsymbol{\mu}_i$ is equilibrium level of the latent state toward which the system is pulled over time.

Continuous-Time Dynamic Structure:

The continuous-time parameterization, when `ct = TRUE`, for the dynamic structure is given by

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\alpha}_i + \boldsymbol{\beta}_i \boldsymbol{\eta}_{i,t-1}) dt + \boldsymbol{\Psi}_i^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

note that $d\mathbf{W}$ is a Wiener process or Brownian motion, which represents random fluctuations.

If `center = TRUE`, the dynamic structure is parameterized as follows

$$d\boldsymbol{\eta}_{i,t} = \boldsymbol{\beta}_i (\boldsymbol{\eta}_{i,t-1} - \boldsymbol{\mu}_i) dt + \boldsymbol{\Psi}_i^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

Value

Returns an object of class `varmxid` which is a list with the following elements:

call Function call.

args List of function arguments.

fun Function used ("FitVARMxID").

output A list of fitted OpenMx models.

robust A list of output from `OpenMx::mxRobustSE()` with argument `details = TRUE` for each id if `robust = TRUE`.

Author(s)

Ivan Jacob Agaloos Pesigan

References

Hunter, M. D. (2017). State space modeling in an open source, modular, structural equation modeling environment. *Structural Equation Modeling: A Multidisciplinary Journal*, 25(2), 307–324. doi:10.1080/10705511.2017.1369354

Neale, M. C., Hunter, M. D., Pritikin, J. N., Zahery, M., Brick, T. R., Kirkpatrick, R. M., Estabrook, R., Bates, T. C., Maes, H. H., & Boker, S. M. (2015). OpenMx 2.0: Extended structural equation and statistical modeling. *Psychometrika*, 81(2), 535–549. doi:10.1007/s1133601494358

See Also

Other VAR Functions: `LDL()`, `Softplus()`

Examples

```
# Generate data using the simStateSpace package-----
library(simStateSpace)
set.seed(42)
n <- 5
time <- 100
p <- 2
alpha <- rep(x = 0, times = p)
beta <- 0.50 * diag(p)
psi <- 0.001 * diag(p)
```

```

psi_l <- t(chol(psi))
mu0 <- simStateSpace::SSMMeanEta(
  beta = beta,
  alpha = alpha
)
sigma0 <- simStateSpace::SSMCovEta(
  beta = beta,
  psi = psi
)
sigma0_l <- t(chol(sigma0))
sim <- SimSSMVARFixed(
  n = n,
  time = time,
  mu0 = mu0,
  sigma0_l = sigma0_l,
  alpha = alpha,
  beta = beta,
  psi_l = psi_l
)
data <- as.data.frame(sim)

# Fit the model-----
# center = TRUE
library(fitVARmxD)
fit <- FitVARmxD(
  data = data,
  observed = paste0("y", seq_len(p)),
  id = "id",
  center = TRUE
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)
converged(fit)

# Fit the model-----
# center = FALSE
library(fitVARmxD)
fit <- FitVARmxD(
  data = data,
  observed = paste0("y", seq_len(p)),
  id = "id",
  center = FALSE
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)
converged(fit)

```

LDL

*LDL' Decomposition of a Symmetric Positive-Definite Matrix***Description**

Performs an LDL' factorization of a symmetric positive-definite matrix X , such that

$$X = LDL^{\top},$$

where L is unit lower-triangular (ones on the diagonal) and D is diagonal.

Usage

```
LDL(x, epsilon = 1e-10)
```

```
InvLDL(s_l, uc_d)
```

Arguments

<code>x</code>	Numeric matrix. Must be symmetric positive-definite.
<code>epsilon</code>	Numeric. Small positive value used to replace zero diagonal entries.
<code>s_l</code>	Matrix. Strictly lower-triangular part of L .
<code>uc_d</code>	Vector. Unconstrained vector with $\log_{1p}(\exp(uc_d)) = d$.

Details

`LDL()` returns both the unit lower-triangular factor L and the diagonal factor D . The strictly lower-triangular part of L is also provided for convenience. The function additionally computes an unconstrained vector `uc_d` such that `softplus(uc_d) = d`, using $\text{softplus}^{-1}(y) = \log(\exp(y) + 1)$ for stable back-transformation. `InvLDL()` returns a symmetric positive definite matrix from the strictly lower-triangular part of L and the unconstrained vector `uc_d`.

Value

- `LDL()`: a list with components:
 - `l`: a unit lower-triangular matrix L
 - `s_l`: a strictly lower-triangular part of L
 - `d`: a vector of diagonal entries of D
 - `uc_d`: unconstrained vector with `softplus(uc_d) = d`
 - `x`: input matrix
- `InvLDL()`: a symmetric positive definite matrix

See Also

Other VAR Functions: [FitVARMxID\(\)](#), [Softplus\(\)](#)

Examples

```

set.seed(123)
x <- crossprod(matrix(rnorm(16), 4, 4)) + diag(1e-6, 4)
ldl <- LDL(x = x)
ldl
inv_ldl <- InvLDL(s_l = ldl$s_l, uc_d = ldl$uc_d)
inv_ldl
max(abs(x - inv_ldl))

```

print.varmxid

Print Method for Object of Class varmxid

Description

Print Method for Object of Class varmxid

Usage

```

## S3 method for class 'varmxid'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  var_metric = "var",
  converged = TRUE,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  prop = FALSE,
  grad_tol = 0.01,
  hess_tol_abs = 1e-08,
  hess_tol_rel = 1e-10,
  check_condition = FALSE,
  cond_max = 1e+12,
  abs_bnd_tol = 1e-06,
  rel_bnd_tol = 1e-04,
  ok_codes = 0L,
  require_finite_fit = TRUE,
  digits = 4,
  ...
)

```

Arguments

<code>x</code>	an object of class <code>varmxid</code> .
<code>means</code>	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
<code>mu</code>	Logical. If <code>mu = TRUE</code> , include estimates of the <code>mu</code> vector, if available. If <code>mu = FALSE</code> , exclude estimates of the <code>mu</code> vector.
<code>alpha</code>	Logical. If <code>alpha = TRUE</code> , include estimates of the <code>alpha</code> vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the <code>alpha</code> vector.
<code>beta</code>	Logical. If <code>beta = TRUE</code> , include estimates of the <code>beta</code> matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the <code>beta</code> matrix.
<code>nu</code>	Logical. If <code>nu = TRUE</code> , include estimates of the <code>nu</code> vector, if available. If <code>nu = FALSE</code> , exclude estimates of the <code>nu</code> vector.
<code>psi</code>	Logical. If <code>psi = TRUE</code> , include estimates of the <code>psi</code> matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the <code>psi</code> matrix.
<code>theta</code>	Logical. If <code>theta = TRUE</code> , include estimates of the <code>theta</code> matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the <code>theta</code> matrix.
<code>var_metric</code>	Character string. If <code>var_metric = "var"</code> , <code>psi</code> and <code>theta</code> are in the original variance/covariance metric. If <code>var_metric = "logvar"</code> , the diagonal elements of <code>psi</code> and <code>theta</code> are the log of the variances and the off-diagonal elements correspond to strict <code>L</code> in the <code>LDL'</code> decomposition. If <code>var_metric = "softplusvar"</code> , the diagonal elements of <code>psi</code> and <code>theta</code> are the softplus of the variances and the off-diagonal elements correspond to strict <code>L</code> in the <code>LDL'</code> decomposition.
<code>converged</code>	Logical. Only include converged cases.
<code>vanishing_theta</code>	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
<code>theta_tol</code>	Numeric. Tolerance for vanishing <code>theta</code> test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
<code>prop</code>	Logical. If <code>prop = FALSE</code> , a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If <code>prop = TRUE</code> , the proportion of cases that converged.
<code>grad_tol</code>	Numeric. Tolerance for the gradient-based convergence check.
<code>hess_tol_abs</code>	Numeric. Absolute tolerance used when checking whether the Hessian is positive-definite.
<code>hess_tol_rel</code>	Numeric. Relative tolerance used when checking whether the Hessian is positive-definite.
<code>check_condition</code>	Logical. If <code>TRUE</code> , also check the Hessian condition number as part of the Hessian diagnostics.
<code>cond_max</code>	Numeric. Maximum allowed condition number for the Hessian when <code>check_condition = TRUE</code> .
<code>abs_bnd_tol</code>	Numeric. Absolute tolerance for detecting whether a parameter is at its lower or upper bound.

<code>rel_bnd_tol</code>	Numeric. Relative tolerance for detecting whether a parameter is at its lower or upper bound.
<code>ok_codes</code>	Integer vector. Optimizer status codes that are considered acceptable.
<code>require_finite_fit</code>	Logical. If TRUE, require a finite objective value (fit) as part of the convergence check.
<code>digits</code>	Integer indicating the number of decimal places to display.
<code>...</code>	further arguments.

Value

Prints means or raw estimates depending the the value of the argument means.

Author(s)

Ivan Jacob Agaloos Pesigan

Softplus

Softplus and Inverse Softplus Transformations

Description

The softplus transformation maps unconstrained real values to the positive real line. This is useful when parameters (e.g., variances) must be strictly positive. The inverse softplus transformation recovers the unconstrained value from a positive input.

Usage

`Softplus(x)`

`InvSoftplus(x)`

Arguments

`x` Numeric vector or matrix. Input values to be transformed.

Details

- $\text{Softplus}(x) = \log(1 + \exp(x))$
- $\text{InvSoftplus}(x) = \log(\exp(x) - 1)$

For numerical stability, these functions use `log1p()` and `expm1()` internally.

Value

- `Softplus()`: numeric vector or matrix of strictly positive values.
- `InvSoftplus()`: numeric vector or matrix of unconstrained values.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other VAR Functions: [FitVARMxID\(\)](#), [LDL\(\)](#)

Examples

```
# Apply softplus to unconstrained values
x <- c(-5, 0, 5)
y <- Softplus(x)

# Recover unconstrained values
x_recovered <- InvSoftplus(y)

y
x_recovered
```

summary.varmxid

Summary Method for Object of Class varmxid

Description

Summary Method for Object of Class varmxid

Usage

```
## S3 method for class 'varmxid'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  var_metric = "var",
  converged = TRUE,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  prop = FALSE,
  grad_tol = 0.01,
  hess_tol_abs = 1e-08,
  hess_tol_rel = 1e-10,
  check_condition = FALSE,
```

```

    cond_max = 1e+12,
    abs_bnd_tol = 1e-06,
    rel_bnd_tol = 1e-04,
    ok_codes = 0L,
    require_finite_fit = TRUE,
    digits = 4,
    ...
)

```

Arguments

object	an object of class varmxid.
means	Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.
mu	Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
var_metric	Character string. If var_metric = "var", psi and theta are in the original variance/covariance metric. If var_metric = "logvar", the diagonal elements of psi and theta are the log of the variances and the off-diagonal elements correspond to strict L in the LDL' decomposition. If var_metric = "softplusvar", the diagonal elements of psi and theta are the softplus of the variances and the off-diagonal elements correspond to strict L in the LDL' decomposition.
converged	Logical. Only include converged cases.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
prop	Logical. If prop = FALSE, a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If prop = TRUE, the proportion of cases that converged.
grad_tol	Numeric. Tolerance for the gradient-based convergence check.
hess_tol_abs	Numeric. Absolute tolerance used when checking whether the Hessian is positive-definite.

hess_tol_rel	Numeric. Relative tolerance used when checking whether the Hessian is positive-definite.
check_condition	Logical. If TRUE, also check the Hessian condition number as part of the Hessian diagnostics.
cond_max	Numeric. Maximum allowed condition number for the Hessian when check_condition = TRUE.
abs_bnd_tol	Numeric. Absolute tolerance for detecting whether a parameter is at its lower or upper bound.
rel_bnd_tol	Numeric. Relative tolerance for detecting whether a parameter is at its lower or upper bound.
ok_codes	Integer vector. Optimizer status codes that are considered acceptable.
require_finite_fit	Logical. If TRUE, require a finite objective value (fit) as part of the convergence check.
digits	Integer indicating the number of decimal places to display.
...	further arguments.

Value

Returns means or raw estimates depending the the value of the argument means.

Author(s)

Ivan Jacob Agaloos Pesigan

vcov.varmxid

Sampling Covariance Matrix of the Parameter Estimates

Description

Sampling Covariance Matrix of the Parameter Estimates

Usage

```
## S3 method for class 'varmxid'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  var_metric = "var",
```

```

    converged = TRUE,
    vanishing_theta = TRUE,
    theta_tol = 0.001,
    prop = FALSE,
    grad_tol = 0.01,
    hess_tol_abs = 1e-08,
    hess_tol_rel = 1e-10,
    check_condition = FALSE,
    cond_max = 1e+12,
    abs_bnd_tol = 1e-06,
    rel_bnd_tol = 1e-04,
    ok_codes = 0L,
    require_finite_fit = TRUE,
    robust = FALSE,
    ...
)

```

Arguments

object	Object of class varmxid.
mu	Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
var_metric	Character string. If var_metric = "var", psi and theta are in the original variance/covariance metric. If var_metric = "logvar", the diagonal elements of psi and theta are the log of the variances and the off-diagonal elements correspond to strict L in the LDL' decomposition. If var_metric = "softplusvar", the diagonal elements of psi and theta are the softplus of the variances and the off-diagonal elements correspond to strict L in the LDL' decomposition.
converged	Logical. Only include converged cases.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
prop	Logical. If prop = FALSE, a named logical vector indicating, for each individual fit, whether the convergence criteria are met. If prop = TRUE, the proportion of cases that converged.

<code>grad_tol</code>	Numeric. Tolerance for the gradient-based convergence check.
<code>hess_tol_abs</code>	Numeric. Absolute tolerance used when checking whether the Hessian is positive-definite.
<code>hess_tol_rel</code>	Numeric. Relative tolerance used when checking whether the Hessian is positive-definite.
<code>check_condition</code>	Logical. If TRUE, also check the Hessian condition number as part of the Hessian diagnostics.
<code>cond_max</code>	Numeric. Maximum allowed condition number for the Hessian when <code>check_condition</code> = TRUE.
<code>abs_bnd_tol</code>	Numeric. Absolute tolerance for detecting whether a parameter is at its lower or upper bound.
<code>rel_bnd_tol</code>	Numeric. Relative tolerance for detecting whether a parameter is at its lower or upper bound.
<code>ok_codes</code>	Integer vector. Optimizer status codes that are considered acceptable.
<code>require_finite_fit</code>	Logical. If TRUE, require a finite objective value (fit) as part of the convergence check.
<code>robust</code>	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix.
<code>...</code>	additional arguments.

Value

Returns a list of sampling variance-covariance matrices.

Author(s)

Ivan Jacob Agaloos Pesigan

Index

* VAR Functions

FitVARMXID, [5](#)

LDL, [14](#)

Softplus, [17](#)

* fitVARMXID

FitVARMXID, [5](#)

LDL, [14](#)

Softplus, [17](#)

* fit

FitVARMXID, [5](#)

* methods

coef.varmxid, [2](#)

converged, [4](#)

print.varmxid, [15](#)

summary.varmxid, [18](#)

vcov.varmxid, [20](#)

* misc

LDL, [14](#)

Softplus, [17](#)

coef.varmxid, [2](#)

converged, [4](#)

FitVARMXID, [5](#), [14](#), [18](#)

InvLDL (LDL), [14](#)

InvSoftplus (Softplus), [17](#)

LDL, [12](#), [14](#), [18](#)

OpenMx::imxRobustSE(), [12](#)

print.varmxid, [15](#)

Softplus, [12](#), [14](#), [17](#)

summary.varmxid, [18](#)

vcov.varmxid, [20](#)